Evaluation of Climate Hazards Center InfraRed Precipitation with Stations Data

Jackie Chen, Mario Tapia-Pacheco, Peng Zhao, Sarah Liang, Shuai Yuan

Mentor: Erika McPhillips | Sponsor: Shrad Shukla

UC SANTA BARBARA Geography

Climate Hazards Center UC SANTA BARBARA

UC SANTA BARBARA | Data Science Initiative

Abstract

Daily **precipitation (PPT)** information is crucial for monitoring climate extremes. Four versions of daily CHIRPS 3.0 were developed using different supporting data from NOAA, NASA, CCCS, and the CHRS. This project studies the behaviors of these daily CHIRPS 3.0 versions and aims to validate their PPT values. The project included::

- Creating difference maps for initial pairwise comparison
- Utilizing Granger Causality test to examine the predictive relationship of **soil moisture (SM)** and PPT
- Validating PPT by comparing peaks of SM and PPT
- Proving that independent datasets are informative for predicting short term missing values, which contributes to the study of regions with fewer data and information

Data

CHIRPS 3.0 is a quasi-global, daily, gridded PPT dataset. There are four different versions, or "flavors", based on each of the following datasets:

1. **CPC-IR** → National Oceanic and Atmospheric Administration (NOAA)
2. **ERA5** → Copernicus Climate Change Service (CCCS)
3. **PERSIANN-CDR** → Center for Hydrometeorology and Remote Sensing at the UC Irvine (CHRS)
4. **IMERG** → National Aeronautics and Space Administration (NASA)

Daily PPT measurements from these four datasets are rescaled to align their monthly totals with the CHIRPS 2.0 monthly totals. Other datasets used for validation are:

- **PRISM** → United States based PPT
- **SM2RAIN-ASCAT** → simulated PPT, based on SM
  ** SM2RAIN algorithm simulates extremely specific PPT; too specific in dry seasons especially
- **MOD11A1** → land surface temperature (LST)
- **GLEAM** → SM, land evaporation, etc.

All these datasets were collected by separate entities and we were interested in analyzing their relationship with precipitation and whether or not they could be used to validate the different flavors of the CHIRPS 3.0 dataset.

All datasets used are 3D arrays, with dimensions longitude, latitude, and time. Preprocessing prior to analysis included:

- Coarsening datasets with different spatial resolutions
- Subsetting datasets for regional focus and efficient computation
- Managing missing values (identifying, interpolating, etc.)

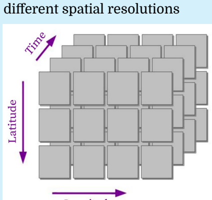

Figure 1: Visual representation of three-dimensional array data.

Methodology

Pairwise Comparisons

- Log mean variance maps
  - Non-negative, uninformative in terms of under/overestimation of PPT
- Rain rate difference maps → **QR code slides 2-4**
  - Informative in under/overestimation

$$d_{\text{PPT}} = \frac{\text{number of days} > 5mm}{\text{number of total days}}$$

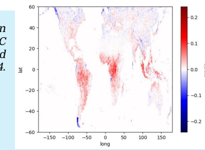

Figure 2: Difference in rain rate between CPC and IMERG based CHIRPS 3.0 in 2014.

Assessing Extreme Values

- Granger Causality test
  - Statistical hypothesis test → can one time series be used to forecast another?
  - Using GLEAM SM data and CHIRPS 3.0 PPT; expectation of PPT causing SM
  - Extreme values in PPT are correlated with extreme values in SM.
- Consider two time series that are **correlated**, generate time series of their **extreme values**, and then assess the **similarity** of these two newly-generated time series.

Figure 3: Workflow of extreme values validation; thresholds, lag, similarity metric, etc. need to be considered.

SARIMA Modeling

- Comparing SARIMA components of CHIRPS 3.0 versions vs. SM2RAIN-ASCAT

Validation using Independent PPT Datasets

- U.S. and global log mean variance and daily rain rate difference maps (CHIRPS 3.0 versions vs. **SM2RAIN-ASCAT** and **PRISM)**
- Flagging wet and dry day disagreements (CHIRPS 3.0 versions vs. **SM2RAIN-ASCAT)** → **QR code slide 7-8**

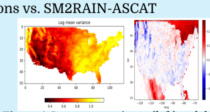

Figure 4: Log mean variance (left) and daily rain rate difference (right) between IMERG and PRISM PPT in the U.S. in 2008

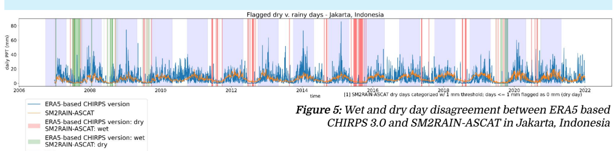

Figure 5: Wet and dry day disagreement between ERA5 based CHIRPS 3.0 and SM2RAIN-ASCAT in Jakarta, Indonesia

Generating PPT from Land Parameters with Conditional GAN

- Learn a mapping from land parameters (e.g. soil moisture) to PPT
- Leverage the completeness of satellite data to fill regions of uncertainty in station data

- "Image to image" problem (Pix2Pix framework)
  - Input 5-day window of land parameters
  - Generator output 1-day PPT
  - Discriminator output real/fake classification
  - Weighted L1 similarity loss + discriminator loss
- Model training and the effects of two losses → **QR code slides 13-15**

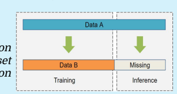

Figure 6: Intuition of cross-dataset generation

- Can land parameters act as natural rain gauges and can we extract the information of PPT from them?

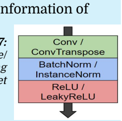

Figure 7: Downsample/ upsample building blocks of G/D net

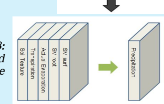

Figure 8: Input and output shape

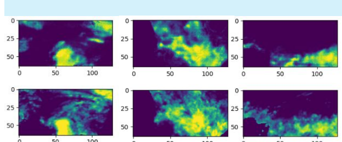

Figure 9: (PPT) Target image (2nd row) and generated image (1st row)

Findings and Results

Pairwise Comparison Conclusion:

- Largest variance and rain rate differences in tropical regions (Amazon rainforest, central **Africa**, **Southeast Asia**)
- (underestimating) PERSIANN < CPC ≈ IMERG < ERA5 (overestimating) → **QR code slides 2-4**

Extreme Values Conclusion:

- Provides a seemingly viable method for validation using independent, non-precipitation datasets
- This method requires a more interpretable similarity metric
  - Euclidean distance and symbolic aggregate approximation provide good similarity metrics, but are not necessarily interpretable on their own
- Granger causality test on PPT vs SM shows suggested causation in regions with less precipitation,
  - Causation reflects the strongest with a lag of 2-4 days.
  - SM dataset slight agrees more with IMERG, but differences between the four flavors are not very significant.
  - PPT has a stronger causation with SM2Rain compared to SM. → **QR code slides 18-20**

SARIMA Modeling Conclusion:

- In Kuala Lumpur, Malaysia, the ERA5 based CHIRPS 3.0 and SM2RAIN-ASCAT data follow similar behavior modeled by a similar SARIMA(1,0,0)(0,1,1)[365] model → **QR code slides 5-6**
- Difficult to apply in areas with regular 0 mm dry seasons and volatile wet seasons (heavily non-stationary areas)

Validation using Independent PPT Datasets Conclusion:

Global **log mean variance** (SM2RAIN-ASCAT vs. CHIRPS 3.0 versions) → **QR code slide 9**
- Lowest log mean variance in arid, desert climates (North Africa, Arabian Peninsula)
- Highest log mean variance in tropical climates (Melanesia, West Papua, Papua New Guinea), some South America)

Global daily **rain rate difference** map *(difference = CHIRPS 3.0 – SM2RAIN-ASCAT)* → **QR code slide 9**
- Arid climates → SM2RAIN-ASCAT ≈ CHIRPS 3.0 versions
- Coastal areas → SM2RAIN-ASCAT < CHIRPS 3.0 versions *(coastal portion of the Eastern United States, the southwestern coast of South America, most coastal areas of Southeast Asia, the northwestern coast of North America, most of Canada, the central portion of Africa, and some desert areas)*
- Tropical climates → SM2RAIN-ASCAT > CHIRPS 3.0 versions *(most of South America (besides the southern portion) and Africa in addition to South and most of Southeast Asia (aside from the coastal areas of Southeast Asia))*

Validation with focus on arid, coastal, tropical regions

- PRISM: entire continental United States
- SM2RAIN-ASCAT: eight California cities

United States **log mean variance** (PRISM vs. CHIRPS 3.0 versions) → **QR code slide 10 and Figure 4**
- Higher log mean variance in coastal Northwest and several areas of the Eastern United States like the Northeast
- Lower log mean variance in Southwestern United States

United States daily **rain rate difference** *(difference = CHIRPS 3.0 versions – PRISM)* → **QR code slide 10 and Figure 4**
- CHIRPS 3.0 versions < PRISM → Western United States (particularly the coastal Northwest)
- CHIRPS 3.0 versions > PRISM → portions of Midwest (including Wisconsin and Michigan) and Eastern (including parts of New England and coastal areas) United States

California and tropical Southeast Asia **daily rain rate difference** *(difference = CHIRPS 3.0 – SM2RAIN-ASCAT)*
- Daily rainfall in SM2RAIN-ASCAT for California cities seemed to be more in line with the PPT from the four versions compared to the Southeast Asian cities → **QR code slide 11**
- Differences between SM2RAIN-ASCAT and the four versions for the California cities were generally less than the differences for the Southeast Asian cities → **QR code slide 12**

Southeast Asia wet and dry day disagreements (ERA5 based CHIRPS 3.0 version vs. SM2RAIN-ASCAT)
- At a 1 mm threshold to cut off fine accuracy of SM2RAIN, ERA5 based CHIRPS 3.0 version follows the recordings of wet and dry days well, expect for anomalies like 2015 El Nino affecting Jakarta (Figure 5), etc. → **QR slides 7-8**

PPT Generation Conclusion:

- 5-day window of soil moisture (and other land parameters) does contain a lot of information about 1-day PPT, as attempts to recover the PPT from SM yield decent visual results
- The flexible framework of CGAN can be adapted to a variety of geographical data generation tasks

Future Work

- Implement current validation techniques on other possibly correlated datasets such as land surface temperature
- Find a more interpretable similarity metric for extreme value validation
- Research other possible validation methods for CHIRPS 3.0 dataset
- Focus on the regions and dates where validation failed, use some quantitative method to estimate the value
- Adopt metrics to validate the resulted fake PPT and document a complete generated dataset

References and Acknowledgements

Brocca, L., L. Ciabatta, C. Massari, T. Moramarco, S. Hahn, S. Hasenauer, R. Kidd, W. Dorigo, W. Wagner, and V. Levizzani (2014), Soil as a natural rain gauge: Estimating global rainfall from satellite soil moisture data, *J. Geophys. Res. Atmos.*, 119, 5128–5141, doi:10.1002/2014JD021489.

Funk, C., Peterson, P., Landsfeld, M. et al. The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes. Sci Data 2, 150066 (2015). https://doi.org/10.1038/sdata.2015.66

P. Isola, J. -Y. Zhu, T. Zhou and A. A. Efros, "Image-to-Image Translation with Conditional Adversarial Networks," 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 2017, pp. 5967-5976, doi: 10.1109/CVPR.2017.632.

Granger, C. W. J. "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods." Econometrica, vol. 37, no. 3, 1969, pp. 424–38. JSTOR, https://doi.org/10.2307/1912791. Accessed 29 May 2024.

Daly, Christopher, et al. "Physiographically sensitive mapping of climatological temperature and precipitation across the conterminous United States." International Journal of Climatology, vol. 28, no. 15, 12 Mar. 2008, pp. 2031–2064, https://doi.org/10.1002/joc.1688.

Martens, B., Miralles, D. G., Lievens, H., van der Schalie, R., de Jeu, R. A. M., Fernández-Prieto, D., Beck, H. E., Dorigo, W. A., and Verhoest, N. E. C.: GLEAM v3: satellite-based land evaporation and root-zone soil moisture, Geosci. Model Dev., 10, 1903–1925, https://doi.org/10.5194/gmd-10-1903-2017, 2017.

Goodfellow, Ian, et al. "Generative Adversarial Networks." Communications of the ACM, vol. 63, no. 11, 22 Oct. 2020, pp. 139–144, https://doi.org/10.1145/3422622.

Soil moisture data: https://www.gleam.eu/
Land surface temperature data: https://modis.gsfc.nasa.gov/data/dataprod/mod11.php
PRISM data: https://prism.oregonstate.edu/
SM2RAIN-ASCAT data: https://zenodo.org/records/10376109